

Voylla Project Architecture

1. Overview

Voylla is a modular, extensible Python tool for automated test case generation using AI. It supports multiple AI providers (OpenAI, Anthropic, Mock) and is designed for easy integration, teaching, and hackathon use.

2. High-Level Architecture

- **CLI Entry Point:**
 - [main.py](#)
 - Handles command-line arguments, loads configuration, and orchestrates the workflow.
- **Configuration Management:**
 - [ai_config.py](#), [configuration_manager.py](#)
 - Loads settings from [.env](#) and YAML files, manages API keys, model selection, and feature flags.
- **Provider Factories:**
 - [ai_provider_factory.py](#)
 - Instantiates the correct AI provider (OpenAI, Anthropic, or Mock) based on configuration.
- **AI Providers:**
 - Implementations for each supported AI service, encapsulating API calls and response handling.
- **Analyzers:**
 - [code_analyzer.py](#), [function_analyzer.py](#), [code_parser.py](#)
 - Parse and analyze user code to extract functions, classes, and edge cases.
- **Generators:**
 - [test_generator.py](#)
 - Generate test cases using AI provider outputs and code analysis results.
- **Integrations:**
 - [github_actions.py](#)
 - (Planned) CI/CD and GitHub integration for automated test generation in workflows.

- **Interfaces:**
 - [base_interfaces.py](#)
 - Defines abstract base classes for configuration, providers, analyzers, and generators.
- **Testing:**
 - [demo_tests](#), test_output/
 - Contains unit and integration tests, as well as generated test outputs.

3. Data Flow

1. **User runs CLI** →
2. **Configuration loaded** from [.env](#)/YAML →
3. **AI Provider selected** via factory →
4. **Code analyzed** for structure and edge cases →
5. **Test cases generated** using AI or mock →
6. **Tests output** to files or stdout.

4. Extensibility

- Add new AI providers by implementing the provider interface and updating the factory.
- Add new analyzers or generators by following the interface contracts.
- Integrate with CI/CD by extending the integrations module.

5. Key Files and Folders

- [src](#) — Main source code
- [main.py](#) — CLI entry point
- [config](#) — Configuration management
- [factories](#) — Provider and generator factories
- [analyzers](#) — Code analysis modules
- [generators](#) — Test generation logic
- [integrations](#) — CI/CD and external integrations
- [interfaces](#) — Abstract interfaces
- [demo_tests](#) — Example and unit tests
- test_output/ — Generated test files

6. Configuration

- [.env](#) — Main environment variables (provider, model, API keys, etc.)
- [config.yaml](#) — Optional advanced configuration

7. Supported Providers

- **OpenAI** (gpt-3.5-turbo, gpt-4, etc.)
- **Anthropic** (Claude models)
- **Mock** (for demos, no API required)